DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #04C: WEAPONS ARSENAL (PART 3 — SCHOLARLY, MUSICAL & DHARMA CONDUIT FAMILIES: FOCUS, ZITHER, BRUSH, ROSARY)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Weapon Audit Series Plan (`04A–04E`)**:
  - `04A`: Group 1 — Core Bladed & Polearm Melee (`SWORD`, `BLADE`, `SPEAR`, `STAFF`)
  - `04B`: Group 2 — Brawling, Ballistic & Flexible (`FIST`, `CLAW`, `BOW`, `NEEDLE`, `WHIP`)
  - **`04C` (This Document)**: Group 3 — **Scholarly, Musical & Dharma Conduits (`FOCUS` ของวิเศษสื่อปราณ/ลูกแก้ว/ขนนกยูงกลไก, `ZITHER` พิณเจ็ดสาย/ขลุ่ย/ระฆัง, `BRUSH` พู่กันอาคม/สิ่วค่ายกล, `ROSARY` ลูกประคำบารมีธรรม)**
  - `04D`: Group 4 — Grand Dao Tactical Implements (`FAN`, `ASTROLABE`, `ABACUS`)
  - `04E`: Part 5 — Holistic Weapon Ecosystem & Cross-System Catch-All Audit

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0. Hard Architectural Invariants for AI Implementers

1. **Immutable `human_core.js`**: Never modify `GAME_SOURCE/js/human_core.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`). Note `FOCUS` lives in `h.skills.FOCUS`, whereas `ZITHER`, `BRUSH`, and `ROSARY` route through `M.skillRead` / `M.skillAdd`.
2. **Ecological & Crafting Realism**: Animal furs and python tendons must **never** be mined out of rock veins or harvested from herb flowerbeds (`economy.js:354–355`!).
3. **100% Emergent NPC Autonomy**: Zen/Karma monks must recognize `ROSARY` as their sacred implement, `INK_LITERATI` calligraphers must favor `BRUSH`, and `SONIC_HARMONY` virtuosos must favor `ZITHER`.

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1. `[WPN-ECOLOGY-01]` Cross-Family Crafting Material Absurdity: Mining Weasel Fur From Solid Rock & Growing Python Tendons in Herb Beds (`js/economy.js:354–355`)!

Before examining each conduit weapon individually, inspect `js/economy.js:354–355`:

```javascript
const HERBS = {
  1: ['MAT_HERB_DEW_GRASS_1', 'MAT_HERB_CINDER_LEAF_1',
'MAT_ZITHER_PARASOL_WOOD_1', 'MAT_WHIP_IRON_SILKWORM_THREAD_1', 'MAT_V230_002'],
  3: ['MAT_HERB_BLOOD_GINSENG_3', 'MAT_HERB_VENOM_VINE_3',
'MAT_HERB_ANCIENT_MIST_FERN_3', 'MAT_HERB_MERIDIAN_FLOWER_3',
      'MAT_ZITHER_ICE_SILK_STRING_3', 'MAT_WHIP_SHADOW_PYTHON_TENDON_3',
'MAT_V230_010', 'MAT_V230_014']
};
const ORES = {
  1: ['MAT_ORE_BLACK_IRON_1', 'MAT_ORE_CINNABAR_1',
'MAT_BRUSH_SPIRIT_WEASEL_HAIR_1', 'MAT_V230_001', 'MAT_V230_004'], ...
};
```

- **Two Severe Contextual & Economic Ripple Bugs**:
    1. **Absurdity**: Miners swinging pickaxes in the underground ore mine (`ECO.mineYield`) literally dig up **`MAT_BRUSH_SPIRIT_WEASEL_HAIR_1` (ขนหางพังพอนวิญญาณ — Spirit Weasel Tail Hair for Brushes!)** out of rock veins, while herbalists tending medicine beds (`ECO.harvestHerb`) harvest **`MAT_WHIP_SHADOW_PYTHON_TENDON_3` (เส้นเอ็นเหลือมเงา — Python Tendons!)** and **`MAT_WHIP_IRON_SILKWORM_THREAD_1` (ใยหนอนไหมเหล็ก)** as if snake tendons sprout from soil!
    2. **38% Alchemy Herb Dilution Cascade**: Because `2/5` (`40%`) of `HERBS[1]` and `2/8` (`25%`) of `HERBS[3]` are non-herb Zither/Whip weapon parts, **real medicinal herb production in `herb_garden` drops by ~35%**, starving `A.brewBasic` and `ALCHEMY` pill production across long simulations while clogging the warehouse with unused Zither strings!
- **Fix**: Keep `HERBS` strictly for botanical plants/woods (`MAT_ZITHER_PARASOL_WOOD_1` can drop from `grove`/forestry or low-weight arbor plots) and `ORES` strictly for minerals/jade; source animal-based weapon materials (`MAT_BRUSH_SPIRIT_WEASEL_HAIR_1`, `MAT_WHIP_SHADOW_PYTHON_TENDON_3`, `MAT_ZITHER_ICE_SILK_STRING_3`, `MAT_WHIP_IRON_SILKWORM_THREAD_1`) from `wildlife.js` hunts, `beast_menagerie` / `SILK_BEE_KEEPER` husbandry, and traveling merchants!

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2. [WPN-FOCUS-02] Weapon Family #10: FOCUS (ของวิเศษสื่อปราณ / ลูกแก้ว / กระจกสำริด / ขนนกยูงกลไก)

2.1 Naming & Classification Collision With FAN (economy.js:586 & data_items.js)

- **PROCEDURAL_POOLS.baseForms.FOCUS (economy.js:586)**: FOCUS: ['พัดหยกสื่อปราณ', 'กระจกสำริดส่องปราณ', 'ลูกแก้วดารา', 'ป้ายหยกบัญชาปราณ', 'ตราประทับวิญญาณ']
  - **20% (1/5) of procedural FOCUS items are named 'พัดหยกสื่อปราณ' (Jade Fan)** while carrying gearType: 'FOCUS'! When a CELESTIAL_WEATHER Fan master equips it, every FAN skill suffers a **×0.65 mismatch penalty**!
  - Meanwhile, ITEM190_EQ_008 ('พัดหยกดาราดูดกลืนปราณ' in data_items.js) lacks an explicit gearType field!
  - **Fix**: Replace 'พัดหยกสื่อปราณ' in FOCUS baseForms with 'แส้ปัดขนจามรีเซียน' (*Daoist Horsetail Whisk* — 拂尘, iconic XIAN Focus!) and explicitly set gearType: 'FAN' on ITEM190_EQ_008.

2.2 Legacy weaponTypes: ['FOCUS'] Over-Locking Dedicated Implements (data_tech.js)

- Pre-v2.0.0 techniques for Fans, Flutes, Brushes, Needles, Rosaries, Luopans, and Abacuses still hardcode weaponTypes: ['FOCUS'] alone—penalizing disciples who upgrade from a generic FOCUS orb to their path's signature FAN, ZITHER, BRUSH, NEEDLE, ROSARY, ASTROLABE, or ABACUS! Audit data_tech.js and append the matching dedicated gearType alongside 'FOCUS'.

2.3 LEG_WPN_PEACOCK_PLUME_FOCUS (ขนนกยูงทองคำกลไกสวรรค์ — 孔雀翎) Missing Mechanism Paths & Benchmark Ring Bug

- In visitors_clues.js:244–273, *Peacock Plume* is a masterwork mechanical hidden-weapon cylinder (craft: 'ARRAY'), yet **MECH_PUPPET** (CELESTIAL_BALLISTA, ASTRAL_CLOCKWORK_PRIME, PUPPET_RELIC_ARTIFICER) and **NINPO_SHADOW.CONCEALED_WEAPON_STRIKER** are absent from compatiblePath and preferredSubs!
- In VC.benchmarkWeaponBalance (visitors_clues.js:2115), FOCUS is benchmarked against **'MP_RING_5' (a Ring accessory!)** instead of 'MP_FOCUS_5'.

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3. [WPN-ZITHER-03] Weapon Family #11: ZITHER (พิณเจ็ดสาย / ขลุ่ยหยก / ระฆังสำริด / สังข์ปราณ)

3.1 SONIC_HARMONY & SEVEN_EMOTIONS_ZITHER Missing From SUB_EQUIP_BONUSES & LEG_WPN_JIUXIAO_PHOENIX_ZITHER!

- **Locations**: `js/martial.js:469-471`, `js/visitors_clues.js:284-285`
- **Verified Stale IDs & Path Omissions**:
  1. `M.SUB_EQUIP_BONUSES` (`martial.js:470`) keys on **`'SOUL_MUSICIAN'`**—an old renamed ID (`M.normalize` converts it to `'SEVEN_EMOTIONS_ZITHER'`)! As a result, `HEART_SOUL.SEVEN_EMOTIONS_ZITHER` and **all 6 subpaths of `SONIC_HARMONY` (`SEVEN_STRINGS_ANNIHILATION`, `JADE_FLUTE_SOUL_CHARMER`, `WAR_DRUM_TEMPEST_HERALD`, `RESONANT_BELL_EXORCIST`, `SILENT_WAVE_ASSASSIN`, `CELESTIAL_CHOIR_CONDUCTOR`)** fall back to generic bonuses!
  2. `LEG_WPN_JIUXIAO_PHOENIX_ZITHER` (`visitors_clues.js:284-285`) sets `compatiblePath: ['XIAN', 'HEART_SOUL', 'SAGE_SUPPORT', 'ANY']`—**forgetting `'SONIC_HARMONY'`**, and its `preferredSubs` lists stale `'SOUL_MUSICIAN'` and nonexistent `'CHANT_RESONANCE'` while omitting `SEVEN_EMOTIONS_ZITHER` and all 6 `SONIC_HARMONY` subpaths (costing them `+135` suitability on the ultimate Phoenix Zither)!

3.2 Musical Instrument Subtypes (Guqin vs Flute vs War-Drum/Bell) & Tea/Pavilion Harmony

- Differentiate `ZITHER` sub-instruments via `visualSubtype` / name tags:
  - `ZITHER` / `FLUTE` (พิณ / ขลุ่ย): `RANGED_KITE` (`baseRange: 2.75`, `+cast` / `+mdef`).
  - `BELL` / `DRUM` (ระฆังสำริด / กลองศึก — `WAR_DRUM_TEMPEST_HERALD` / `RESONANT_BELL_EXORCIST`): `MID_TACTICAL` AoE acoustic pulse (`+hp` / `+poise`, grants nearby allies `fear`/`confusion` resistance).
  - Out-of-combat: Equipping a T4+ `ZITHER` boosts `harmonic_pavilion` / `tea` social stress-relief and Heart-Demon soothing by `+15%`!

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4. [WPN-BRUSH-04] Weapon Family #12: BRUSH (พู่กันเทวะ / พู่กันจารึกยันต์ / สิ่วสลักค่ายกล)

4.1 Triple Live Combat Bug in LEG_WPN_CHUNQIU_JUDGMENT_BRUSH (visitors_clues.js:304–307 vs 865–875)!

- **Exact Code (visitors_clues.js:305–306 vs 865–875)**:
  - Spec (lines 305–306): descTh: 'ทะลวงเกราะป้องกันวิชา (+18% ความเสียหายวิชา) และผนึกการเคลื่อนไหวเป้าหมาย' params: { bonusDmgMult: 0.18, armorReductionRatio: 0.22, targetStatus: 'armorbreak', selfStatus: 'focusUp', dur: 7, icdTicks: 8 }
  - Handler (lines 865–875):

```JavaScript
} else if (eid === 'EFX_LEG_CHUNQIU_VERDICT_SEAL') {
  attacker.wpnProcCds[eid] = W.t + icd;
  if (COMBAT.alive(defender)) {
    ST.addStatus(defender, p.targetStatus || 'armorbreak', p.dur || 7);
    if (strikeResult && strikeResult.dmg > 0) { ... }
  }
  attacker.lastWeaponProc = ...;
}
```

- **Three Verified Bugs**:
  1. **Applies Physical 'armorbreak' (-def) Instead of Magical 'mdefbreak' (-mdef) + Movement Seal ('root' / 'slow')!**: CHUNQIU_JUDGMENT_BRUSH is an **860 matk Qi/Calligraphy weapon** whose tooltip explicitly promises *"ทะลวงเกราะป้องกันวิชา (mdefbreak) และผนึกการเคลื่อนไหวเป้าหมาย"*—yet it applies only physical 'armorbreak' and **zero** movement restriction!
  2. **Completely Drops selfStatus: 'focusUp'**: Lines 865–875 never apply p.selfStatus to attacker at all!
  3. **Drops armorReductionRatio: 0.22**.
  - **Fix**: Apply 'mdefbreak' + 'slow' (or brief 2-tick 'root' with boss immunity) to defender, and grant 'qiConduit' / 'focusUp' (+cast / +acc) to attacker!

4.2 INK_LITERATI & LORE_ARCHIVIST Omitted From CHUNQIU_JUDGMENT_BRUSH & SUB_EQUIP_BONUSES!

- LEG_WPN_CHUNQIU_JUDGMENT_BRUSH (visitors_clues.js:299–300) omits **INK_LITERATI (วิถีอักษรสวรรค์ — the primary Brush Path!)** and **LORE_ARCHIVIST**,

while referencing stale/deleted subpath IDs `'GRAND_FORMATION_ARCHITECT'`, `'GRAND_STRATEGIST'`, and `'UNDERWORLD_DECREE'`!

- Add `INK_LITERATI` (all 6 subpaths), `LORE_ARCHIVIST` (all 4 subpaths), `EARTH_VEIN_WARDEN`, `BATTLEFIELD_STRATEGIST`, and `SAGE_SUPPORT.CELESTIAL_INK_CHRONICLER` to `CHUNQIU_JUDGMENT_BRUSH` and `M.SUB_EQUIP_BONUSES`, plus grant `+10%` `TALISMAN` / `ARRAY` inscription quality and `annotateHeritageSutra` speed when wielding a T4+ `BRUSH`!

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5. `[WPN-ROSARY-05]` Weapon Family #13: `ROSARY` (ลูกประคำบารมีธรรมร้อยแปดเม็ด)

5.1 Major Path-Profile Disconnect: `KARMA_SAMSARA`, `ZEN_VOID_MIND` & `VAJRA_BODHI` Treat `ROSARY` as an Unfamiliar Weapon (`0.72×`)!

- **Locations**: `js/martial.js:23, 39, 50, 54, 67–72`
- **Verified Contradiction**:
  - `KARMA_SAMSARA`'s canonical path lore explicitly states: **"ทรัพยากรหลัก: ลูกประคำชะตากรรม (`Karmic Rosary Beads`)"**, and `HEART_SOUL.ZEN_VOID_MIND`, `BODY.VAJRA_BODHI`, `GOD.MEDICINE_BODHI_GOD`, and `GOD.GOLDEN_ARHAT_STAFF` are Buddhist/Arhat monastic lineages!
  - **Yet in `js/martial.js` (`paths` & `profiles`), NOT A SINGLE ONE OF THEM INCLUDES `'ROSARY'` IN THEIR PREFERRED WEAPONS** (`KARMA_SAMSARA` lists `['WHIP','BRUSH','STAFF','FOCUS','BLADE','SWORD']`; `ZEN_VOID_MIND`, `LOTUS_MERCY_PURIFIER`, `SIX_PATHS_WHEEL`, `BODHI_VOW_GUARDIAN` list `['STAFF','FOCUS']`)!
  - Consequently, every `KARMA_SAMSARA` and `ZEN_VOID_MIND` monk scores `ROSARY` weapons at **`fit = 0.72 ("อาวุธไม่ถนัด")`** and refuses to equip them!
  - **Fix**: Add `'ROSARY'` to `M.PATHS.KARMA_SAMSARA.weapons`, `M.PATHS.HEART_SOUL.weapons`, and the preferred weapon arrays (`pr[3]`) + `SUB_EQUIP_BONUSES` of `SIX_PATHS_WHEEL`, `LOTUS_MERCY_PURIFIER`, `BODHI_VOW_GUARDIAN`, `CAUSE_EFFECT_LEDGER`, `GOLDEN_WHEEL_ARBITER`, `ZEN_VOID_MIND`, `VOID_LOTUS_CONTEMPLATOR`, `VAJRA_BODHI`, `MEDICINE_BODHI_GOD`, and `GOLDEN_ARHAT_STAFF` (and add them to `LEG_WPN_BODHI_NIRVANA_ROSARY.preferredSubs`)!

5.2 `LEG_WPN_BODHI_NIRVANA_ROSARY`: Zero Clue Quest, Missing `cleanseHeartDemon`, & Slot Disambiguation

1. **Zero Clue Template in `VC.CLUE_TEMPLATES`**: Author `CLUE_LEG_WPN_BODHI_NIRVANA_ROSARY` (เจดีย์พระธาตุโบราณและมหาประคำโพธิญาณร้อยแปดเม็ด — `SCRIPTURE` sutra contemplation + `personality.composure` / nonlethal mercy trial).
2. **Missing Heart-Demon Cleanse in `runWeaponProcs` (`visitors_clues.js:346` vs `876–904`)**:
   - Tooltip promises "ชำระล้างมารในใจ", yet `params` (`line 347`) omits `cleanseHeartDemon: 5` and the shared handler (`lines 876–904`) never reduces `attacker.h.state.heartDemon`! Add `cleanseHeartDemon: 5` to `params` and process it in the `876–904` handler (and route `hpRestorePct` through `COMBAT.healUnit` so `healBlock` is respected!).
3. **Weapon `ROSARY` vs Accessory `NECKLACE` Beads (`ประคำ`) Slot Guard**:
   - Both `WEAPON` (`ลูกประคำ... ROSARY`) and `ACCESSORY` (`ITEM190_EQ_016` ประคำโพธิ์ขาว, `LEG_ACC_BODHI_STARLIGHT_BEADS` ประคำโพธิ์ดารา `NECKLACE`) use `"ประคำ"` in `nameTh`. Ensure `JD.MARTIAL.classify(slot, nameTh)` strictly branches on `slot === 'WEAPON'` vs `slot === 'ACCESSORY'` first so Prayer-Bead Necklaces never misclassify as `ROSARY` weapons (and vice-versa)!

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6. Implementation & Verification Checklist for GPT / Gemini (`04C`)

- [ ] **Fix Crafting Ecology (`js/economy.js:354–355`)**: Remove animal furs/tendons/silkworm threads from `ORES` and `HERBS` so mining and herb farming yield authentic minerals and medicinal herbs without starving `ALCHEMY`.
- [ ] **`FOCUS`**: Replace `'พัดหยกสื่อปราณ'` in `baseForms.FOCUS` with `'แส้ปัดขนจามรีเซียน'`; tag `ITEM190_EQ_008` as `FAN`; append dedicated weapon types to legacy `weaponTypes: ['FOCUS']` techniques; add `MECH_PUPPET` & `CONCEALED_WEAPON_STRIKER` to `PEACOCK_PLUME_FOCUS`; fix `benchmarkWeaponBalance 'MP_FOCUS_5'`.
- [ ] **`ZITHER`**: Replace stale `'SOUL_MUSICIAN'`/`'CHANT_RESONANCE'` with `'SEVEN_EMOTIONS_ZITHER'` and add `SONIC_HARMONY` (all 6 subpaths) to `M.SUB_EQUIP_BONUSES` and `JIUXIAO_PHOENIX_ZITHER`; add `+15% harmonic_pavilion` / `tea` calming aura.
- [ ] **`BRUSH`**: Fix `EFX_LEG_CHUNQIU_VERDICT_SEAL` (`visitors_clues.js:865–875`) to apply `'mdefbreak'` + movement slow/seal and self `'focusUp'`/`'qiConduit'`; add `INK_LITERATI` & `LORE_ARCHIVIST` (removing stale IDs) to `CHUNQIU_JUDGMENT_BRUSH` and `SUB_EQUIP_BONUSES`.

- [ ] **ROSARY**: Add `'ROSARY'` to `KARMA_SAMSARA` & monastic subpaths in `martial.js`; add `cleanseHeartDemon: 5` handling to `EFX_LEG_BODHI_NIRVANA_ROSARY`; author `CLUE_LEG_WPN_BODHI_NIRVANA_ROSARY`; enforce strict `slot`-first guard in `D.classify` for `"ประคำ"`.